1. Create a database and use it

```
mysql> create database school;
Query OK, 1 row affected (0.02 sec)
```

---

2. Create a table using different datatypes

```
CREATE TABLE student (
 student_id INT,
name VARCHAR(50),
 age INT,
fees DECIMAL(8,2),
join_date DATE
);
mysql> create table student(student_id int, name varchar(20), age int, fees decimal(8,2), join_date
date);
Query OK, 0 rows affected (0.11 sec)
```

---

3. Add a new column - email VARCHAR(50);

```
mysql> create table student(student_id int, name varchar(20), age int, fees decimal(8,2), join_date
date);

Query OK, 0 rows affected (0.11 sec)

mysql> desc student;

+------------+--------------+------+-----+---------+-------+

| Field      | Type         | Null | Key | Default | Extra |

+------------+--------------+------+-----+---------+-------+

| student_id | int          | YES  |     | NULL    |       |

| name       | varchar(20)  | YES  |     | NULL    |       |

| age        | int          | YES  |     | NULL    |       |

| fees       | decimal(8,2) | YES  |     | NULL    |       |

| join_date  | date         | YES  |     | NULL    |       |

| email      | varchar(20)  | YES  |     | NULL    |       |

+------------+--------------+------+-----+---------+-------+
```

6 rows in set (0.01 sec)

---

4. Modify datatype - age TINYINT;

mysql> alter table student modify age tinyint;

Query OK, 0 rows affected (0.16 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student;

+------------+--------------+------+-----+---------+-------+

| Field | Type | Null | Key | Default | Extra |

+------------+--------------+------+-----+---------+-------+

| student_id | int | YES | | NULL | |

| name | varchar(20) | YES | | NULL | |

| age | tinyint | YES | | NULL | |

| fees | decimal(8,2) | YES | | NULL | |

| join_date | date | YES | | NULL | |

| email | varchar(20) | YES | | NULL | |

+------------+--------------+------+-----+---------+-------+

6 rows in set (0.01 sec)

---

5. Drop a column – email.

mysql> alter table student drop column email;

Query OK, 0 rows affected (0.20 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student;

```
+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
| student_id | int          | YES  |     | NULL    |       |
| name       | varchar(20)  | YES  |     | NULL    |       |
| age        | tinyint      | YES  |     | NULL    |       |
| fees       | decimal(8,2) | YES  |     | NULL    |       |
| join_date  | date         | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+
```

5 rows in set (0.00 sec)

---

6. Drop table -> Student

```
mysql> drop table student;
```

Query OK, 0 rows affected (0.08 sec)

---

7. Create a table using different datatypes

```
1. CREATE TABLE student (
2. student_id INT,
3. name VARCHAR(50),
4. age INT,
5. fees DECIMAL(8,2),
6. join_date DATE
7. );
```

```
mysql> create table student(student_id int, name varchar(50), age int, fees decimal(8,2),
join_date date);
Query OK, 0 rows affected (0.06 sec)

mysql> desc student;
+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
```

```
+------------+--------------+------+-----+---------+-------+
| student_id | int          | YES  |     | NULL    |       |
| name       | varchar(50)  | YES  |     | NULL    |       |
| age        | int          | YES  |     | NULL    |       |
| fees       | decimal(8,2) | YES  |     | NULL    |       |
| join_date  | date         | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

---

8. Insert single record

Insert multiple records – at list 10 rows

(2, 'Anita', 21, 26000.00, '2024-06-02'),
(3, 'Amit', 19, 24000.75, '2024-06-03');

```
mysql> insert into student(student_id, name, age, fees, join_date) values (2, 'Anita', 21, 26000.00,
'2024-06-02'),
    -> (3, 'Amit', 19, 24000.75, '2024-06-03'),
    -> (4, 'sonal', 19, 25000.75, '2026-06-03'),
    -> (5, 'sakshi', 29, 29000.75, '2026-08-03'),
    -> (6, 'kanishka', 31, 45000.75, '2026-07-03'),
    -> (7, 'poonam', 31, 67000.75, '2026-03-03'),
    -> (8, 'kanam', 31, 67000.75, '2026-03-03'),
    -> (9, 'sanam', 31, 67000.75, '2026-03-03'),
    -> (10, 'sami', 31, 67000.75, '2026-03-03');
Query OK, 9 rows affected (0.06 sec)
Records: 9  Duplicates: 0  Warnings: 0

mysql> select * from student;
+------------+----------+------+----------+------------+
| student_id | name     | age  | fees     | join_date  |
+------------+----------+------+----------+------------+
|          2 | Anita    |   21 | 26000.00 | 2024-06-02 |
|          3 | Amit     |   19 | 24000.75 | 2024-06-03 |
|          4 | sonal    |   19 | 25000.75 | 2026-06-03 |
|          5 | sakshi   |   29 | 29000.75 | 2026-08-03 |
|          6 | kanishka |   31 | 45000.75 | 2026-07-03 |
|          7 | poonam   |   31 | 67000.75 | 2026-03-03 |
|          8 | kanam    |   31 | 67000.75 | 2026-03-03 |
|          9 | sanam    |   31 | 67000.75 | 2026-03-03 |
|         10 | sami     |   31 | 67000.75 | 2026-03-03 |
+------------+----------+------+----------+------------+
9 rows in set (0.00 sec)
```

---

9. Update data from students

fees = 27000
WHERE student_id = 2.

```
mysql> update student set fees=2700 where student_id=2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from student;
+------------+----------+------+----------+------------+
| student_id | name     | age  | fees     | join_date  |
+------------+----------+------+----------+------------+
|          2 | Anita    |   21 |  2700.00 | 2024-06-02 |
|          3 | Amit     |   19 | 24000.75 | 2024-06-03 |
|          4 | sonal    |   19 | 25000.75 | 2026-06-03 |
|          5 | sakshi   |   29 | 29000.75 | 2026-08-03 |
|          6 | kanishka |   31 | 45000.75 | 2026-07-03 |
|          7 | poonam   |   31 | 67000.75 | 2026-03-03 |
|          8 | kanam    |   31 | 67000.75 | 2026-03-03 |
|          9 | sanam    |   31 | 67000.75 | 2026-03-03 |
|         10 | sami     |   31 | 67000.75 | 2026-03-03 |
+------------+----------+------+----------+------------+
9 rows in set (0.00 sec)
```

---

10. Delete a record

WHERE student_id = 3

```
mysql> delete from student where student_id=3;

Query OK, 1 row affected (0.01 sec)

mysql> select * from student;

+------------+----------+------+----------+------------+

| student_id | name     | age  | fees     | join_date  |

+------------+----------+------+----------+------------+

|          2 | Anita    |   21 |  2700.00 | 2024-06-02 |

|          4 | sonal    |   19 | 25000.75 | 2026-06-03 |

|          5 | sakshi   |   29 | 29000.75 | 2026-08-03 |

|          6 | kanishka |   31 | 45000.75 | 2026-07-03 |
```

| 7 | poonam | 31 | 67000.75 | 2026-03-03 |

| 8 | kanam | 31 | 67000.75 | 2026-03-03 |

| 9 | sanam | 31 | 67000.75 | 2026-03-03 |

| 10 | sami | 31 | 67000.75 | 2026-03-03 |

+------------+----------+------+----------+------------+

8 rows in set (0.00 sec)

---

11. Which datatype is used for names?

Ans= varchar

12. Which datatype is best for money?

Ans= decimal

---

13. CREATE TABLE attendance

1. in_time DATETIME,
2. out_time TIME

mysql> create table attendence(in_time datetime, out_time time);
Query OK, 0 rows affected (0.08 sec)

mysql> desc attendence;
+----------+----------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+----------+----------+------+-----+---------+-------+
| in_time | datetime | YES | | NULL | |
| out_time | time | YES | | NULL | |
+----------+----------+------+-----+---------+-------+
2 rows in set (0.01 sec)

---

14. Create a database → college

Create a table Student on this database and insert data at list 5 row

```
mysql> create database college;
Query OK, 1 row affected (0.05 sec)
mysql> use college;
Database changed
mysql> create table student(student_id int, name varchar(20), age int);
Query OK, 0 rows affected (0.14 sec)

mysql> insert into student(student_id, name, age)
    -> values (1, 'Sonal', 22),
    -> (2, 'sakshi', 23),
    -> (3, 'shweta', 24),
    -> (4, 'sami', 25),
    -> (5, 'sani', 26);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

---

15. Create a Course table with PRIMARY KEY

course_id INT PRIMARY KEY,

course_name VARCHAR(40)

```
mysql> create table course(course_id int primary key, course_name varchar(40));
Query OK, 0 rows affected (0.18 sec)

mysql> desc course;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
```

```
+-------------+-------------+------+-----+---------+-------+
| course_id   | int         | NO   | PRI | NULL    |       |
| course_name | varchar(40) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
2 rows in set (0.05 sec)
```

---

16. Add a new column to table → email

```
mysql> alter table course add email varchar(20);
Query OK, 0 rows affected (0.13 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc course;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| course_id   | int         | NO   | PRI | NULL    |       |
| course_name | varchar(40) | YES  |     | NULL    |       |
| email       | varchar(20) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

---

17. Rename a column → course_name

```
mysql> alter table course rename column course_name to c_name;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc course;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| course_id | int         | NO   | PRI | NULL    |       |
| c_name    | varchar(40) | YES  |     | NULL    |       |
| email     | varchar(20) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

---

18. Drop a column → email

```
mysql> alter table course drop column email;
Query OK, 0 rows affected (0.14 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc course;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| course_id | int         | NO   | PRI | NULL    |       |
| c_name    | varchar(40) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
```

2 rows in set (0.03 sec)

---

19. Truncate a table – Student

mysql> TRUNCATE TABLE student;

Query OK, 0 rows affected (0.11 sec)

mysql> SELECT * FROM student;

Empty set (0.00 sec)

---

20. Drop a table → student

mysql> drop table student;

Query OK, 0 rows affected (0.04 sec)
